

RIBBON-design-addendum

RIBBON — Design Addendum: Base Requirements · Colorways · Fold-Out

Rev C companion to the build spec and business plan

1 · Base requirements (formalized)

The structural base is now specified as numbered requirements a fabricator can be held to.

#	Requirement	Value / rule
B1	Structure	Single central steel spine (200 × 100 × 6 mm RHS equivalent), top cantilevered; no leg, apron, or brace in any knee-clear bay
B2	Feet	Two weighted pedestals under the standing zones; base plates ≥700 × 700 × 12 mm with internal ballast
B3	Anti-tip	Withstand a 350 N horizontal pull at any top edge and a 100 kg point load on the edge (child sitting/climbing) without tipping
B4	Load	≥150 kg/m ² distributed; deflection ≤3 mm at any cantilevered end under load
B5	Knee clearance	Four bays: surface 740 mm, ≥685 mm clear underside , ≥480 mm depth, ≥760 mm width — preserved in every configuration and fold state
B6	Leveling	M16 adjustable glides; level on floors up to 10 mm out of plane
B7	Footrail	42 mm 304-stainless tube at the tall zones, 200 mm above floor
B8	Safety	No pinch points, no exposed fasteners in child reach, all corners R20+ bullnose
B9	Finish	Powder-coat (colorway-selectable, see §2), low-VOC, BIFMA-relevant durability
B10	Serviceability	Base disassembles with standard tools for flat-pack shipping; every part serialized to the unit's digital twin

These become acceptance criteria in fabricator contracts and the QA gates in the digital thread.

2 · Colorway system

Color becomes a **configurator dimension** alongside height mix and inserts — three coordinated layers per colorway: top (solid surface), base (powder-coat), and accents (trough inserts, footrail, well tubs).

Launch colorways (working names) | Colorway | Top | Base | Accents | Aimed at | |—|—|—|—| | **Cloud** | Warm white | Deep navy | Teal | Flagship / civic | | **Grove** | Soft sage | Charcoal | Natural oak lids | Museums, libraries | | **Citrus** | Pale sand | White | Coral + amber wells | Children's / education | | **Slate** | Mid grey | Black | Stainless | Coworking / hospitality | | **Custom** | Per spec | RAL any | Per spec | Institutional branding (premium) |

Engineering constraints (real, from solid-surface fabrication practice): - Not all solid-surface colors thermoform — large-particulate/chip lines are excluded; the palette is drawn from **thermoformable** solid-surface ranges only, validated per brand (Corian / HI-MACS / Staron guides). - Light, low-pattern colors hide seams and repairs best — the launch palette biases that way; dark tops are offered with a documented care note. - Powder-coat = any RAL at no premium; **custom top colors** carry a surcharge and minimum-order because they break sheet-purchasing economies. - Colorways change **nothing structural** — same molds/toolpaths, so SKUs multiply at near-zero manufacturing cost. Accent parts (lids, tubs) are the cheap variety layer.

3 · Fold-out functionality

Design intent: fold for **shipping, storage, and reconfiguration** — not a daily one-person fold. The table is ~150+ kg; the win is freight volume, doorway access, and venues that must clear the floor seasonally.

Mechanism — fold at the module joints (tri-fold): - The top already decomposes into height modules + two end caps (Phase 2 architecture). The fold-out version places **two engineered hinge lines at module joints** on the straight runs, splitting the table into a center section + two folding wings. - Hinge lines fall at **flat plateau points of the wave** (never mid-curve), with a machined stainless piano-hinge spine below the surface and a compression-lock latch; the visible joint is a tight seam, not the seamless monolith. - The **central trough splits** with the fold: the stainless liner is made in three gasketed segments; inserts span only within a segment. - Folded envelope: **≈ 1.5 × 1.2 m footprint** (wings vertical) — through a standard double door, and cuts LTL freight class/volume meaningfully. - **Safety:** gas-strut damping on the wings, positive lock engaged/disengaged with a tool (child-safe — B8 applies), no finger gaps >8 mm at the hinge line in any state, and knee-clear bays (B5) preserved when deployed.

Honest trade-offs — offered as a variant, not the default: - Two visible seams interrupt the seamless top (the flagship’s signature); seam quality becomes a QA gate. - ~+\$400-700 unit cost (hinges, locks, segmented liner, added QA) at pilot volumes. - Slightly reduced edge stiffness at the hinge line — B4 deflection must be re-verified per unit. - **Product line:** *RIBBON One* (seamless flagship) and *RIBBON Fold* (tri-fold variant). Fold is expected to win in schools, event spaces, and multi-use civic rooms; One in permanent installations.

4 · Ripple effects across the project

- **Build spec:** base section restated as B1-B10; colorway + Fold variant sections appended.
- **Configurator:** two new dimensions (colorway; One/Fold) → still one parametric file per order.
- **Crossover model:** Fold cuts the freight input (smaller shipped volume) and adds ~\$400-700 unit cost — in freight-dominated regions Fold *improves* the centralized case and slightly delays the open-a-node break-even; test both in the model’s yellow cells.
- **Roadmap:** Fold prototypes in **Phase 1 (pilot)** — at least one pilot site gets a Fold unit; colorways launch with the configurator MVP; custom colorway unlocks at Phase 2.
- **Pilot protocol additions:** fold/unfold cycle count and time, latch/pinch audit, seam wear at the hinge line, and colorway preference intercepts.
- **Business plan:** product section now carries the two-variant line and colorway system; Rev C.

6 · Rev F — Transition pieces (the connective tissue of Ribbon Segment architecture (Level-as-Ribbon))

Transition pieces are now first-class parts: they bridge adjacent ribbon segments so a docked arrangement reads as one flowing system, and they close the safety gaps docking creates.

The family

Part	Code	Function
Straight ramp cap	TC-Δxxx-S	Bridges two spliced segments of height delta Δ (e.g., TC-Δ090-S bridges 90 mm). Locks into both tables’ B12 latches, covers the seam.

Part	Code	Function
Corner ramp cap	TC- Δ xxx-C60 / C90	Same, turning 60° (Hex corners) or 90° (custom layouts).
Ring wedge cap	RC- Δ xxx	Curved transition for Ring wedge-ribbons.
End cap	EC-hhh	Finishes an exposed dock face at height hhh with a bullnosed terminus (open-ended and growing arrangements).
Level bridge	LB-hhh	Zero-delta seam cover joining two same-height tables into one continuous run.

Engineering rules (honest physics)

- **Delta classes:** $\Delta \leq 120$ mm $\rightarrow$ 300 mm ramp; Δ 121–240 mm $\rightarrow$ 450 mm ramp; $\Delta > 240$ mm $\rightarrow$ **stepped cap** (two mini-plateaus), because a single ramp would exceed $\sim 28^\circ$ and anything set on it slides.
- **No-set-down zone:** any cap steeper than 10° carries a molded ridge texture and contrast band — a visual/tactile “don’t park your drink here” cue (and a cane-detectable edge).
- **Safety integration:** caps span the inter-table seam with zero entrapment gaps (B12), add ≤ 1 mm surface step, and cannot be lifted without tool release (child-safe).
- **Material:** same solid-surface family as tops, colorway-matched or deliberately contrast-banded (configurator choice); underside GF-nylon chassis with the latch interface.
- **Count per arrangement:** Racetrack $\approx$ 8–10 caps + 0–2 ECs; Hex $\approx$ 6 corner caps; Ring $\approx$ 8 RCs. Caps are the highest-count SKU in the system — tooled early, injection/compression molded at volume.

Spec impact — B13 (new): transition pieces. Every docked height change carries a cap of the correct delta class; caps meet B12 gap/step limits; $\geq 10^\circ$ caps carry the no-set-down treatment; caps removable only by tool; arrangement is non-compliant if any dock face is left open without an EC.